

Emanuel Munoz

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EDUCATION

Carnegie Mellon University, Robotics Institute

Master of Science in Robotics GPA 4.05

- Relevant coursework: **Embodied Safe AI, Computer Vision, Robot Learning, Multi-agent planning**

Pittsburgh, PA

Expected graduation: August 2026

University of Engineering and Technology, UTEC

Bachelor of Science in Electrical Engineering | GPA: 4.0/4.0

- Thesis: **Active Robotic Exploration via PPO Reinforcement Learning and Self-Supervised Learning**

Lima, Peru

December 2021

EXPERIENCE

Carnegie Mellon University, Robotics Institute

Graduate Research Assistant

- Developed a **real-time safe pipeline** based on **Gaussian Processes, Model Predictive Control**, and **Safe Optimal Control** for drone navigation in smoke-dense environments, mitigating critical visual sensing impairment risks.
- Designed and implemented an **online autoregressive probabilistic predictor** using **Neural Operators** for **behavior forecasting** of highly dynamic smoke, enabling hazard avoidance.
- Engineered a **hierarchical risk-aware path planning** framework for **multi-robot navigation** in hazardous areas, reducing dense smoke exposure by up to ~85% compared to baselines ([Under review IROS 2026](#)).

Pittsburgh, PA

August 2024 – Present

Noteworthy AI

Graduate Machine Learning Engineer Intern

- **Fine-tuned ViT** models using long-tailed datasets, reaching ~90% F1-score in object grid defect classification.
- Prototyped **Out-of-Distribution (OOD)** detectors for ViT models to automatically identify unseen image samples, achieving up to ~70% recall for automated incremental training.

Connecticut, CT | Remote

May 2025 – August 2025

Noteworthy AI

Machine Learning Engineer

- Engineered a **real-time edge inspection** system using **C++** and **ROS2** nodes on **Jetson Orin** to capture localized telemetry and image data from utility poles, enabling automated infrastructure monitoring via fleet vehicles.
- Developed a **stereo visual tracking** pipeline integrating **YOLO detector** and **depth imaging** for **spatial state localization** of target assets, used for physical-to-digital asset correlation.
- Designed a **low-latency** triggering framework using **model predictive tracking** to intercept target assets from moving vehicles, enabling automated, high-resolution industrial camera capture.
- Built cloud-based **ML pipelines** on **AWS** using **PyTorch** and W&B for training, inference, test, and evaluation of production-level ViT models, for asset detection and defect classification.

Connecticut, CT | Remote

March 2022 – August 2024

Carnegie Mellon University, Robotics Institute, RISS Program

Research Scholar

- Designed a **safety-critical motion planning** framework for self-driving vehicles, integrating **barrier functions, optimal control**, and **adaptive learning** to mitigate dynamic uncertainties.
- Validated framework robustness within the CARLA simulation environment, demonstrating zero safety violations under **dynamic uncertainty**; research published at [IROS 2022](#).

Pittsburgh, PA | Remote

May 2021 – July 2022

University of Engineering and Technology, UTEC - CONCYTEC

Research Assistant

- Engineered a feature-based **multi-stereo camera registration** pipeline to reconstruct a surgical environment for teleoperation tasks, delivering **real-time 3D point cloud** representations.
- Developed a **hybrid learning controller** to map torque signals onto position-based manipulators, enabling **full torque controllability** and mitigated tracking errors ([CDC 2020](#)).

Lima, Peru

March 2020 – April 2021

SELECTED PROJECTS

Motion-Token MAPF: Multi-Agent Coordination in Mixed Traffic | [\[GitHub\]](#) | [\[Slides\]](#)

Developed a **multi-agent framework** using an **autoregressive token-based motion predictor** with **Priority-Based Search** to coordinate safe autonomous vehicles amidst human drivers, reducing collision rates and travel times.

PINN-Neural Process: Few-Shot Adaptation Framework for Physics-Informed ML | [\[GitHub\]](#)

Developed a **physics-informed meta-learning framework** with **conditional, variational, and flow matching-based Neural Process** variants to eliminate the retraining bottleneck of standard PINNs, enabling **few-shot adaptation**.

DreamerV2 in ViZDoom: Model-Based Reinforcement Learning | [\[GitHub\]](#)

Implemented **DreamerV2 world model** architecture with a **self-supervised DINO backbone** to evaluate **latent representations** from 3D first-person inputs within the ViZDoom environment for benchmarking.

SKILLS

Programming & ML: Python, PyTorch, Tensorflow, C++, ROS/ROS2, OpenCV.

Deployment & Tools: AWS, Docker, GCP, Weights & Biases, Git, Linux, PostgreSQL, CUDA